

POD AUTOMATION SYSTEM

Product Specification

Technical Reference & Integration Guide

DOCUMENT VERSION	1.0
DATE	March 2026
CLASSIFICATION	Customer Distribution
STANDARD	ISO/IEC 25010 (Software Product Quality)

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Document Version: 1.0 **Classification:** Customer Distribution **Standard:** Aligned with ISO/IEC 25010 (Software Product Quality)

Table of Contents

- 1. [Product Overview](#)
- 2. [System Architecture](#)
- 3. [Technology Stack](#)
- 4. [Core Features & Capabilities](#)
- 5. [System Workflows](#)
- 6. [Integration & API Reference](#)
- 7. [System Requirements](#)
- 8. [Security & Compliance](#)
- 9. [Deployment Architecture](#)
- 10. [Administration & Configuration](#)
- 11. [Performance Characteristics](#)
- 12. [Glossary](#)

1. Product Overview

1.1 Product Name

POD Automation System (POD System v2)

1.2 Purpose

The POD Automation System is a logistics document management platform designed to automate the receipt, classification, matching, and distribution of Proof of Delivery (POD) documents, packing slips, and invoices. It eliminates manual document chasing by integrating email monitoring, AI-powered classification, carrier API connectivity, and a structured approval workflow into a single managed platform.

1.3 Target Users

User Type	Role
Logistics / Operations Teams	Monitor document status, manage carrier requests, review approvals
Accounts / Finance Teams	Retrieve invoices and packing slips linked to orders
System Administrators	Configure the platform, manage users, monitor pipeline health

User Type	Role
External Integrators	Submit and query documents via the REST API using API key authentication

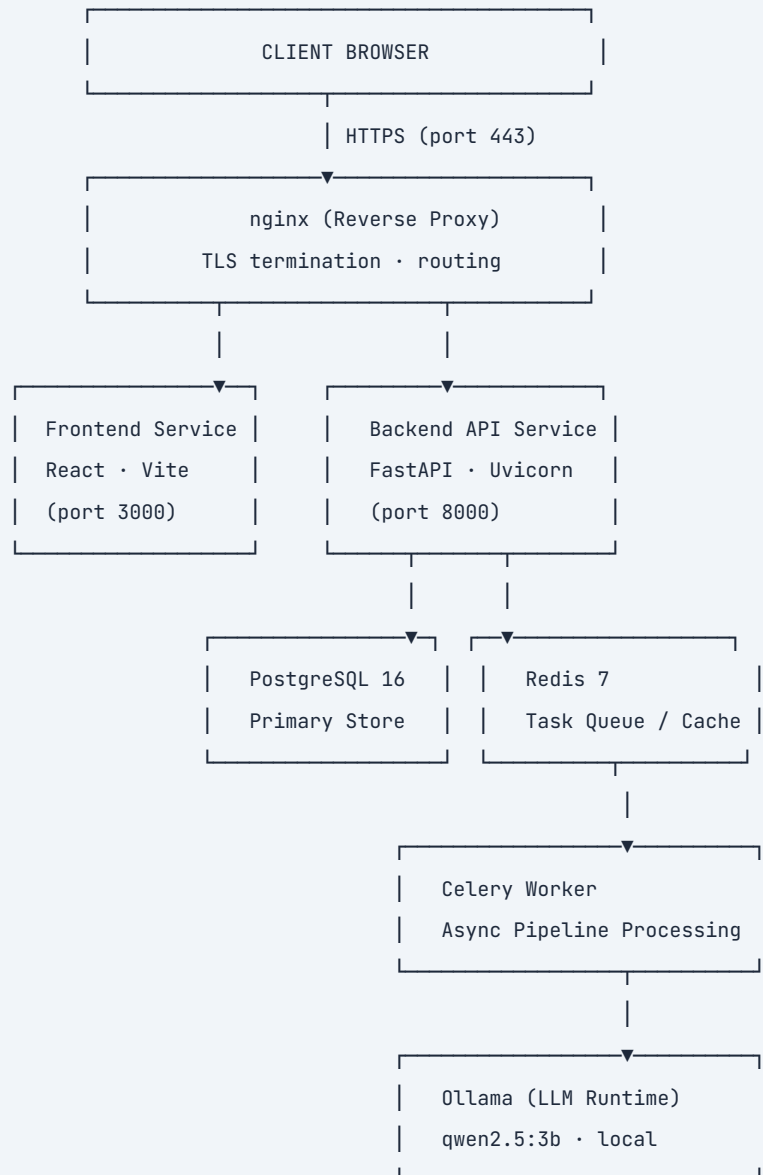
1.4 Key Business Outcomes

- Reduce manual effort in chasing PODs and shipping documents from carriers
- Provide a single searchable record of document status across all deliveries
- Automate email responses to document requests with AI-generated drafts
- Support compliance and audit requirements through full activity logging
- Enable ERP and WMS integration via a documented external REST API

2. System Architecture

2.1 Architectural Overview

The POD System follows a **containerised microservices architecture** deployed via Docker Compose. All components communicate over an internal Docker network; only the reverse proxy is exposed externally.



2.2 Component Responsibilities

Component	Technology	Responsibility
nginx	nginx:alpine	TLS termination, reverse proxy, static asset serving
Frontend	React 18 + Vite 5	Single-page application; all user interaction
Backend API	FastAPI (Python 3.12)	REST API, business logic, authentication
Database	PostgreSQL 16	Persistent storage for all entities
Task Queue	Redis 7 + Celery	Asynchronous pipeline task execution
LLM Runtime	Ollama + qwen2.5:3b	Email classification and response drafting

Component	Technology	Responsibility
IMAP Poller	Background asyncio task	Monitors configured mailboxes for inbound emails

3. Technology Stack

3.1 Backend

Category	Technology	Version
Runtime	Python	3.12
Web Framework	FastAPI	Latest stable
ASGI Server	Uvicorn	Latest stable
ORM	SQLAlchemy (async)	2.x
Database Driver	asyncpg	Latest stable
Task Queue	Celery	Latest stable
Message Broker	Redis	7
Password Hashing	bcrypt	4.1.3
Token Authentication	python-jose (JWT)	Latest stable
HTTP Client	httpx	Latest stable
Email (SMTP)	smtplib (stdlib)	—
Email (IMAP)	imaplib (stdlib)	—
PDF Generation	reportlab	Latest stable
PDF/Image Conversion	Pillow + LibreOffice	Latest stable
Encryption (IMAP creds)	cryptography (Fernet)	Latest stable

3.2 Frontend

Category	Technology	Version
UI Framework	React	18.3
Build Tool	Vite	5.4
Styling	Tailwind CSS	3.4

Category	Technology	Version
Data Fetching	TanStack Query	5.x
HTTP Client	Axios	1.7
Routing	React Router	6.x
Charts	Recharts	2.x
Notifications	Sonner	1.5
Icons	Lucide React	0.454
Spreadsheet I/O	xlsx	0.18
HTML Sanitisation	DOMPurify	3.3

3.3 Infrastructure

Component	Technology	Notes
Container Runtime	Docker + Docker Compose	v2 Compose spec
Reverse Proxy	nginx:alpine	Handles TLS, routing
Database	PostgreSQL 16 Alpine	Persistent volume
LLM Runtime	Ollama	Runs natively on host for GPU access
LLM Model	qwen2.5:3b	~2 GB, local inference
TLS	Self-signed or custom cert	Cert path configurable

4. Core Features & Capabilities

4.1 Document Management

Feature	Description
POD Storage	Stores and indexes Proof of Delivery documents with delivery number, order ID, and carrier linkage
Packing Slip Tracking	Associates packing slips with orders by delivery number or order ID
Invoice Tracking	Associates invoices with orders by invoice number or delivery number
Document Status Dashboard	Single-table view of POD, packing slip, and invoice status per delivery

Feature	Description
Manual Upload (UI)	Drag-and-drop or file-picker upload for all document types
Folder Scanning	Automatic ingestion of documents placed in configured storage folders
FTP Polling	Scheduled retrieval of documents from FTP sources
PDF Conversion	Automatic conversion of images and Office documents to PDF on ingest

4.2 Email Pipeline

Feature	Description
IMAP Monitoring	Polls one or more configured mailboxes at configurable intervals
Subject Filtering	Only emails matching configured subject filters are ingested
AI Classification	LLM classifies email intent (POD request, packing slip, invoice, general) with confidence scoring
Prompt Injection Protection	Pre-screening and output validation prevent malicious email content from manipulating AI behaviour
Multi-Order Detection	Single email referencing multiple order numbers is handled as a multi-order request
Carrier API Lookup	Automatic POD retrieval from UPS, FedEx, and DHL APIs when tracking numbers are available
Unknown Carrier Handling	Automated email request sent to configured carrier contact; request paused awaiting reply
Response Drafting	AI-generated email response drafts with document status table
Approval Workflow	Responses queue for human review and approval before sending
Guidance Queue	Low-confidence classifications or unresolvable orders route to a human review queue
Retry & Recovery	Failed pipeline tasks retry up to 3 times; permanent failures are flagged for manual review

4.3 Order Management

Feature	Description
Order Import	Bulk import via Excel template
Auto-Poll Import	Automatic ingestion of order files placed in a configured import folder
Order CRUD	Full create, read, update, delete via UI
Material Master	Material/SKU reference data management

Feature	Description
Carrier Registry	Carrier records with contact details and API credentials

4.4 User & Access Management

Feature	Description
Role-Based Access	Two roles: Admin (full access) and Reviewer (operational access, no admin functions)
JWT Authentication	Stateless token-based authentication; 4-hour session lifetime
User Management	Admin can create, deactivate, and manage users
API Key Management	Admins can issue, list, and revoke API keys for external integrations
Audit Log	Full activity trail: email events, classification, approvals, document actions, admin operations

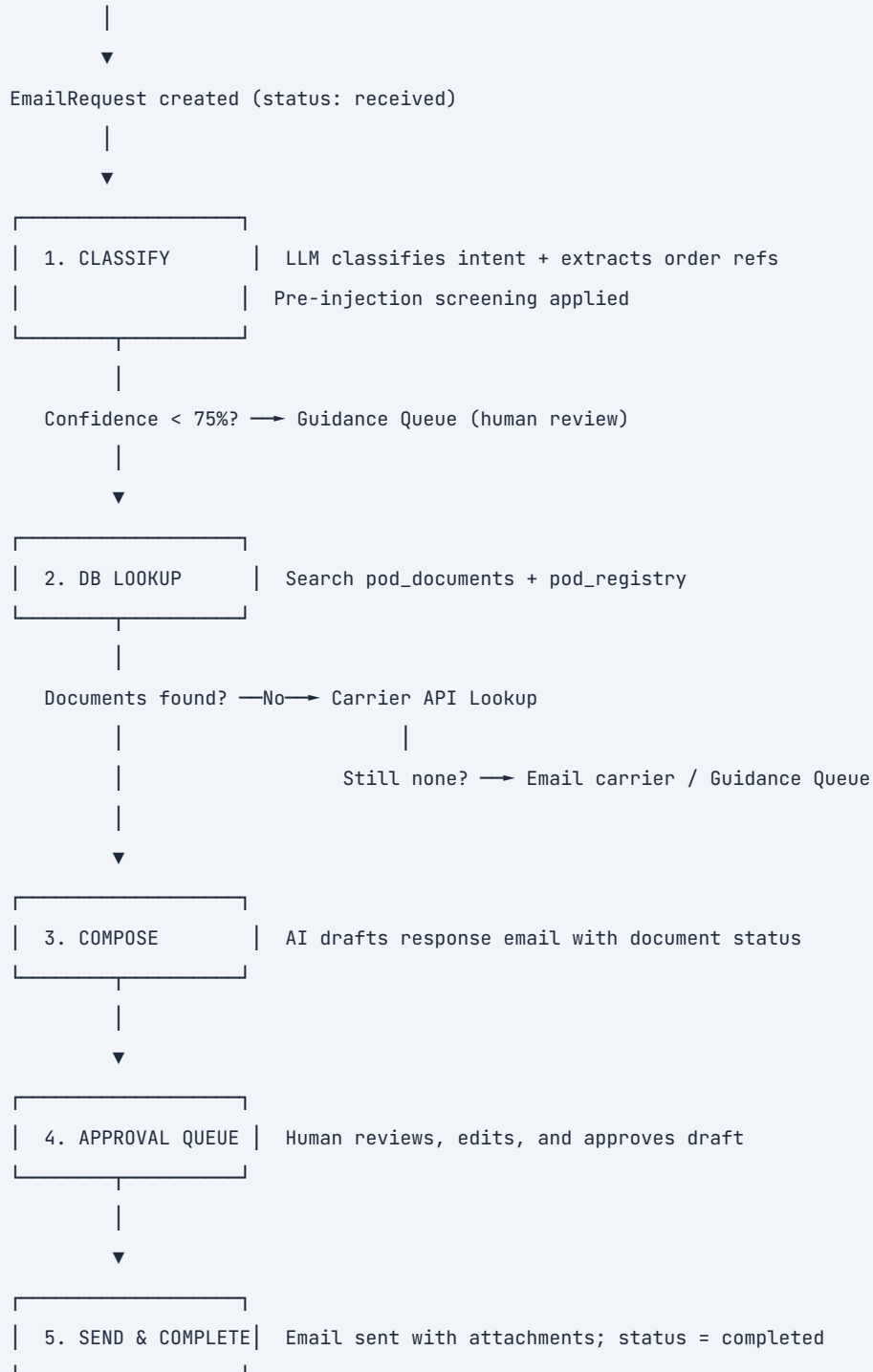
4.5 Reporting & Administration

Feature	Description
Dashboard	Real-time pipeline status chart, request counts by status
Reports Page	Filterable request history with export capability
DB Explorer	Admin-only raw table viewer with paginated data and bulk delete (password-confirmed)
System Configuration	Key-value config store for SMTP, IMAP, carrier credentials, email templates, and system settings
Email Signature	Admin-configurable HTML email signature appended to all outbound emails

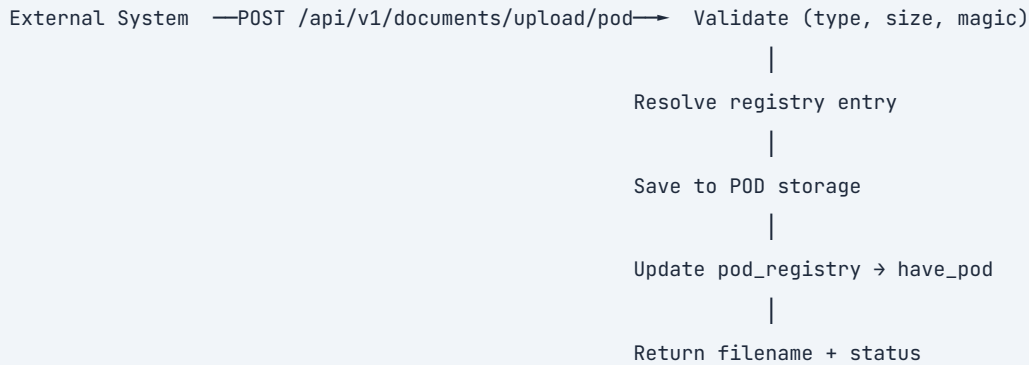
5. System Workflows

5.1 Inbound Email Processing Pipeline

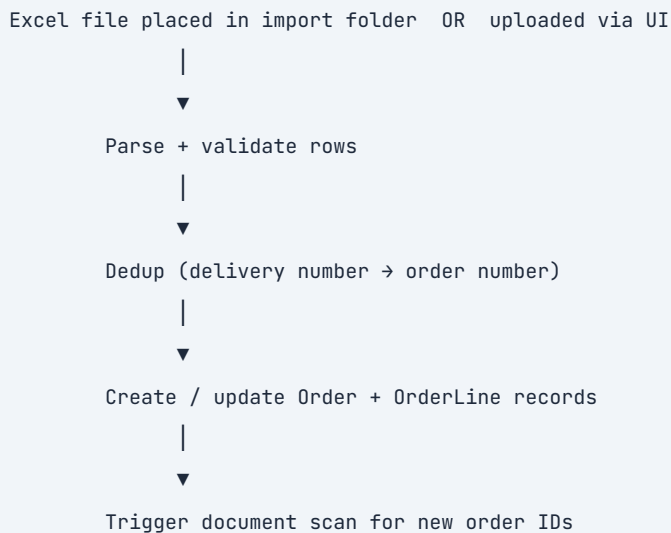
IMAP Poller / Manual Submission



5.2 Document Upload Flow (External API)



5.3 Order Import Flow



6. Integration & API Reference

6.1 External REST API

Base path: `/api/v1/documents/` Authentication: `X-API-Key` header (issued by admin)

Method	Endpoint	Description
GET	<code>/lookup</code>	Query document status by delivery number, customer PO, or order number
POST	<code>/upload/pod</code>	Upload a POD file and associate with a delivery record
GET	<code>/download/pod/{filename}</code>	Download a POD PDF

Method	Endpoint	Description
GET	/download/packing-slip/{filename}	Download a packing slip
GET	/download/invoice/{filename}	Download an invoice

Lookup Query Parameters:

Parameter	Type	Description
delivery_number	string	Internal delivery / despatch number
customer_po	string	Customer purchase order number
order_number	string	Sales or internal order number
request_if_missing	boolean	If <code>true</code> , triggers a carrier POD request when no document is found

Upload Form Fields:

Field	Type	Required	Description
file	file	Yes	PDF or image (PNG, JPEG, TIFF). Max 25 MB.
delivery_number	string	Conditional	At least one identifier required
customer_po	string	Conditional	
order_number	string	Conditional	

6.2 Carrier API Integrations

Carrier	Integration Type	Credentials Required
UPS	REST API (Tracking v3)	Client ID + Client Secret (OAuth 2.0)
FedEx	REST API (Track v1)	Client ID + Client Secret (OAuth 2.0)
DHL	REST API	API Key

Carrier credentials are stored in the system configuration store and are not exposed via any API response.

6.3 Email Integration

Protocol	Purpose	Configuration
SMTP	Outbound email (responses, carrier requests)	Host, port, username, App Password

Protocol	Purpose	Configuration
IMAP	Inbound email monitoring	Host, port, username, encrypted password, SSL

Supported providers: any SMTP/IMAP-compatible provider (verified with Gmail).

7. System Requirements

7.1 Server / Host Requirements

Resource	Minimum	Recommended
CPU	4 cores	8 cores
RAM	8 GB	16 GB
Storage	40 GB	100 GB+ (scales with document volume)
OS	Windows 10/11, Ubuntu 20.04+, macOS 12+	Ubuntu 22.04 LTS
GPU	Not required	DirectML / CUDA (improves LLM inference speed)

Note: The LLM component (`qwen2.5:3b`) requires approximately 2–4 GB of RAM during inference. On systems without a GPU, inference latency is typically 30–120 seconds per classification. A GPU reduces this to 2–10 seconds.

7.2 Software Dependencies (Host)

Software	Version	Purpose
Docker Engine	24.0+	Container runtime
Docker Compose	v2.0+	Service orchestration
Ollama	Latest	Local LLM runtime
Git	Any	Optional — version control

7.3 Network Requirements

Port	Direction	Protocol	Purpose
443	Inbound	HTTPS	Primary application access
80	Inbound	HTTP	Redirects to HTTPS

Port	Direction	Protocol	Purpose
11434	Internal	HTTP	Ollama LLM API (localhost only)
587	Outbound	SMTP/TLS	Email sending
993	Outbound	IMAP/SSL	Email monitoring
443	Outbound	HTTPS	Carrier API calls (UPS, FedEx, DHL)

No inbound ports other than 80 and 443 are required. All backend services communicate on the internal Docker network.

7.4 Browser Requirements

Browser	Minimum Version
Google Chrome	100+
Mozilla Firefox	100+
Microsoft Edge	100+
Safari	15+

JavaScript must be enabled. No browser plugins or extensions are required.

7.5 Storage Sizing Guidance

Data Type	Estimated Size Per Record
POD PDF (generated)	50–200 KB
POD PDF (uploaded)	100 KB – 5 MB
Packing slip / invoice	50 KB – 2 MB
Database record	< 10 KB

A deployment processing 500 deliveries per month with all document types would consume approximately 2–5 GB of storage per year.

8. Security & Compliance

8.1 Authentication & Authorisation

Control	Implementation
User authentication	Username + password with bcrypt hashing (cost factor 12)
Session management	Stateless JWT with 4-hour expiry
Role-based access control	Admin and Reviewer roles with enforced permission boundaries
API authentication	SHA-256 hashed API keys; raw key shown once at creation only
Admin action confirmation	Destructive operations require password re-verification

8.2 Transport Security

Control	Implementation
TLS	HTTPS enforced; HTTP redirects to HTTPS
Certificate	Configurable; self-signed provided for development
Header protection	HSTS, X-Frame-Options, X-Content-Type-Options, Referrer-Policy

8.3 Input & Content Security

Control	Implementation
Prompt injection protection	Pre-screening regex + LLM output validation + input isolation
XSS prevention	DOMPurify sanitisation on all rendered HTML content
File upload validation	Extension check + magic number (binary signature) check + 25 MB limit
SQL injection prevention	Parameterised queries throughout; table names from hardcoded whitelist only
Login brute force protection	10 attempts per 60-second window per IP address

8.4 Credential Handling

Credential	Storage Method
User passwords	bcrypt hash — plaintext never stored
API keys	SHA-256 hash — raw key never stored after creation
IMAP passwords	Fernet symmetric encryption at rest

Credential	Storage Method
JWT secret key	Environment variable (not in source code)
Carrier API credentials	System configuration store (database)

8.5 Audit Logging

All significant system events are recorded in the audit log, including: - Email received and classification outcome - Document lookup and retrieval - Approval and rejection actions - Email send events (including attachment list and delivery status) - Administrative operations (user changes, bulk deletes)

9. Deployment Architecture

9.1 Docker Services

Service Name	Image	Purpose
<code>pod_nginx</code>	nginx:alpine	Reverse proxy, TLS
<code>pod_frontend</code>	Custom (Node 20)	Vite dev server / built assets
<code>pod_backend</code>	Custom (Python 3.12)	FastAPI application
<code>pod_worker</code>	Custom (Python 3.12)	Celery pipeline worker
<code>pod_beat</code>	Custom (Python 3.12)	Celery scheduled task runner
<code>pod_postgres</code>	postgres:16-alpine	Primary database
<code>pod_redis</code>	redis:7-alpine	Task broker and cache

9.2 Persistent Volumes

Volume	Contents
<code>postgres_data</code>	Database files
<code>./pod_storage</code>	POD PDF documents
<code>./packing_slips</code>	Packing slip documents
<code>./invoices</code>	Invoice documents
<code>./order_import</code>	Auto-poll order import folder

9.3 Environment Configuration

All sensitive configuration is supplied via environment variables at deployment time. The following must be set before production deployment:

Variable	Description
<code>SECRET_KEY</code>	JWT signing key — must be a cryptographically random string
<code>ADMIN_SEED_EMAIL</code>	Initial administrator email address
<code>ADMIN_SEED_PASSWORD</code>	Initial administrator password
<code>CORS_ALLOWED_ORIGINS</code>	Comma-separated list of permitted frontend origins
<code>DATABASE_URL</code>	PostgreSQL connection string
<code>REDIS_URL</code>	Redis connection string
<code>OLLAMA_BASE_URL</code>	Ollama API endpoint
<code>OLLAMA_MODEL</code>	LLM model name

10. Administration & Configuration

10.1 System Configuration Keys

Operational settings are managed through the admin Settings interface and stored in the `system_config` table.

Key	Description
<code>smtp_host</code> / <code>smtp_port</code>	Outbound email server settings
<code>smtp_user</code> / <code>smtp_password</code>	SMTP authentication credentials
<code>imap_subject_filters</code>	Comma-separated list of subject keywords for email ingestion
<code>email_signature</code>	HTML signature appended to all outgoing emails
<code>pod_folder_path</code>	Internal path for POD storage
<code>default_pod_request_email</code>	Fallback carrier email for unknown carriers
<code>autopoll_path</code>	Folder path for automatic order file import
<code>fedex_client_id</code> / <code>fedex_client_secret</code>	FedEx API credentials
<code>dhl_api_key</code>	DHL API key

10.2 Common Operational Tasks

Task	Method
Reset pipeline for a stuck request	Requests page → select request → Retrigger
Force immediate IMAP poll	Set <code>last_checked_at = NULL</code> on monitored email record
Purge queued Celery tasks	<code>docker exec pod_worker celery -A app.core.celery_app purge -f</code>
Restart pipeline worker after code change	<code>docker restart pod_worker</code>
Apply configuration changes	Settings → save; backend reloads automatically

11. Performance Characteristics

11.1 Processing Times

Operation	Typical Duration	Notes
Email classification (CPU)	30–120 seconds	Depends on host CPU; no GPU
Email classification (GPU)	2–10 seconds	With compatible GPU via Ollama
Document lookup (DB)	< 100 ms	Indexed queries
API document query	< 200 ms	Including DB lookup
File upload (25 MB)	1–5 seconds	Depends on network and disk speed
PDF conversion (image)	1–3 seconds	Via Pillow
PDF conversion (Office doc)	5–15 seconds	Via LibreOffice headless

11.2 Concurrency

Parameter	Value	Notes
Celery worker concurrency	1	Single-threaded to serialise LLM access
Task retry limit	3 attempts	30-second delay between retries
IMAP poll interval	60 seconds	Configurable per mailbox
Autopoll interval	30 seconds	Fixed

11.3 Scalability Notes

The current deployment is optimised for single-server operation handling hundreds of requests per day. For higher throughput, the worker concurrency can be increased once dedicated GPU resources are available to support parallel LLM inference. The database and API tiers are horizontally scalable independently of the LLM component.

12. Glossary

Term	Definition
POD	Proof of Delivery — a document confirming that goods were delivered and received
Packing Slip	A document listing the contents of a shipment
Pipeline	The automated sequence of steps that processes an inbound email request end-to-end
LLM	Large Language Model — the AI component used for email classification and response drafting
Guidance Queue	A holding queue for requests the system cannot resolve automatically; requires human input
Approval Queue	A queue of AI-drafted email responses awaiting human review before sending
IMAP	Internet Message Access Protocol — used to monitor inbound email mailboxes
SMTP	Simple Mail Transfer Protocol — used to send outbound emails
JWT	JSON Web Token — the authentication token issued on login
API Key	A credential used by external systems to authenticate with the External API v1
Registry	The <code>pod_registry</code> table — the master record of document status per delivery number
Celery	An asynchronous task queue used to process pipeline tasks in the background
Redis	An in-memory data store used as the Celery task broker
Ollama	The local LLM runtime that hosts the classification and drafting model
DOMPurify	A JavaScript library that sanitises HTML content to prevent XSS attacks
Magic Number	The binary signature at the start of a file that identifies its true type

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